College, Debt, and Gender:
The Influence of Outside Options on Higher Education

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Introduction

A seemingly paradoxical **Twin Gender Gap** has emerged in higher education

- Women's matriculation rates are 30% higher than men's
- At the same time, women have worse post-college outcomes
  - 19% lower post-graduate earnings
  - 14% lower labor supply
  - 28% higher student debt balances

Motivating Questions:

- What explains the twin gender gap in higher education?
- What are the implications for gender equality and higher education?

This Paper

Main Mechanism: worse outside options raise net return to college for women.

New data on sources of the twin gender gap in education

- Excess female matriculation from lower income, worse outside option localities
- Disproportionately sort into lower quality, less expensive schools...
- ...but still end up with more student debt and lower earnings.

New implications for gender equality and higher education

→ Gender earnings gap may move in opposite direction of female welfare.

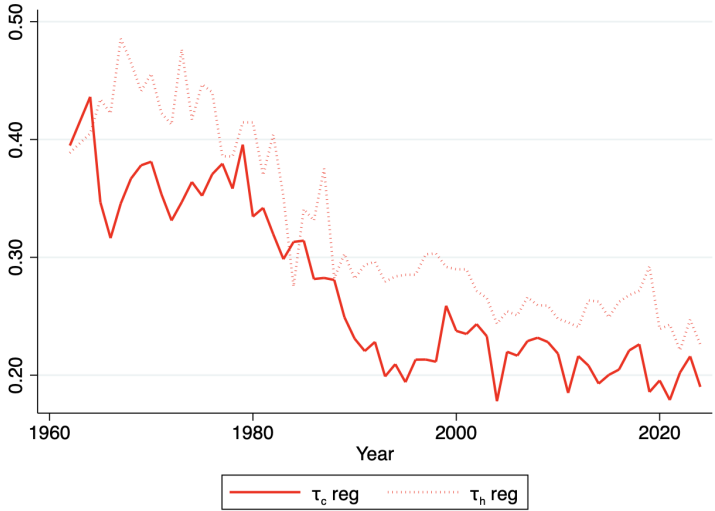
→ Medium term fluctuations in college premium due to compositional effects, unrelated to labor demand.

Today's talk

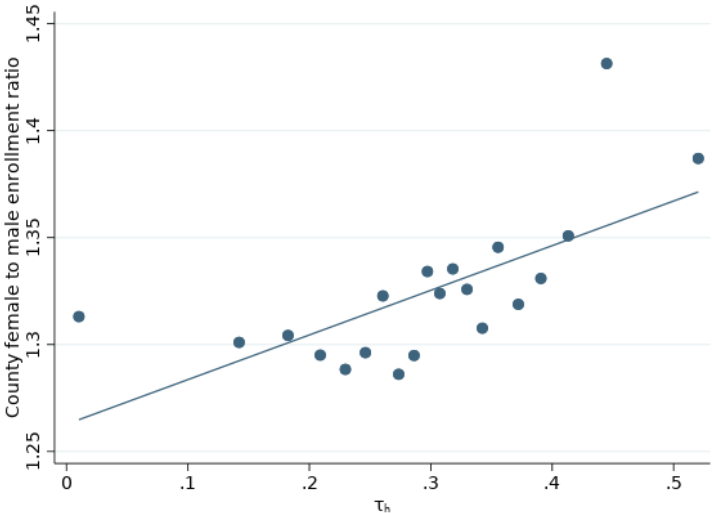
1. Empirical evidence on sources of excess female matriculation
2. Illustrative model
3. Additional facts on college sorting and debt
4. Quantitative model extensions
5. Calibration strategy
6. Preliminary results

Empirical Evidence on Sources of Excess Female Matriculation

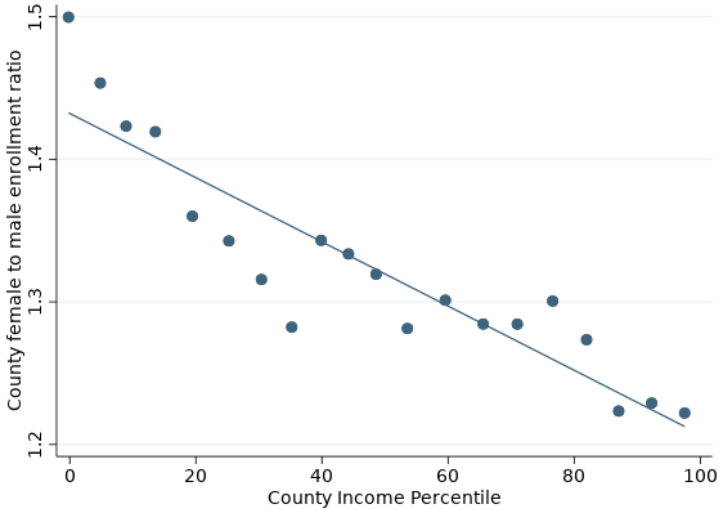
Discrimination in College and Non-College Markets



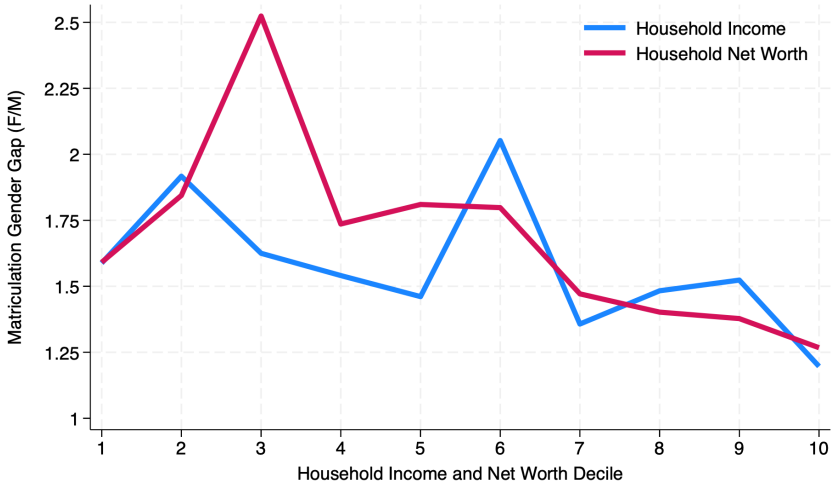
Source of Excess Female Matriculation



Source of Excess Female Matriculation



Source of Excess Female Matriculation



Illustrative Model

An Illustrative Model

High School Graduate.

$$V_g^h = \max\ u(c) \quad \text{s.t.} \quad c \leq w_h(1-\tau_h)$$

College Graduate.

$$V_g^c(\varphi) = \max\ u(c) \quad \text{s.t.} \quad c \leq w_c(1-\tau_c) - \frac{1}{\varphi}p$$

Illustrative Model

Matriculation.

$$\Pi_g \equiv \mathsf{P}\left(\mathsf{V}_g^c(\varphi) > \mathsf{V}_g^h\right)$$

Student Debt.

$$\mathsf{SD}_g \equiv \mathbb{E}\left[\frac{1}{\varphi}\, p \,\middle|\, \text{enrolled in college}\right]$$

Illustrative Model

With heterogeneity in financial support $\varphi \sim \text{Pareto}(\chi, \theta)$

Matriculation.

$$\Pi_g = \left(\chi \left[\frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{p}\right]\right)^{\theta}$$

Student Debt.

$$\text{SD}_g = \frac{\theta}{1+\theta}\left[(1-\tau_c)w_c - (1-\tau_h)w_h\right]$$

Illustrative Model

Matriculation Gap.

$$\frac{\Pi_w}{\Pi_m} = \left[\frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{(w_c - w_h)}\right]^{\theta}$$

Student Debt Gap:

$$\frac{SD_w}{SD_m} = \frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{w_c - w_h}$$

Aggregate Implications

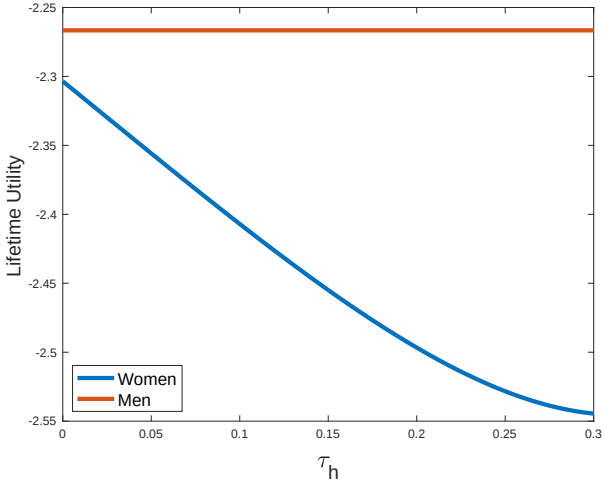
$$\textbf{Agg. Gender Earnings Gap} \quad = \quad \frac{(1-\Pi_w)(1-\tau_h)w_h \;+\; \Pi_w(1-\tau_c)w_c}{(1-\Pi_m)w_h \;+\; \Pi_m w_c}$$

$$\textbf{College Earnings Premium} \quad = \quad \frac{\Pi_w^c(1-\tau_c) \;+\; (1-\Pi_w^c)}{\Pi_w^h(1-\tau_h) \;+\; (1-\Pi_w^h)} \times \frac{w_c}{w_h}$$

where $\quad \Pi_g^c \equiv \dfrac{\Pi_g}{\Pi_w + \Pi_m} \quad , \quad \Pi_g^h \equiv \dfrac{1-\Pi_g}{1-\Pi_w + 1 - \Pi_m}$

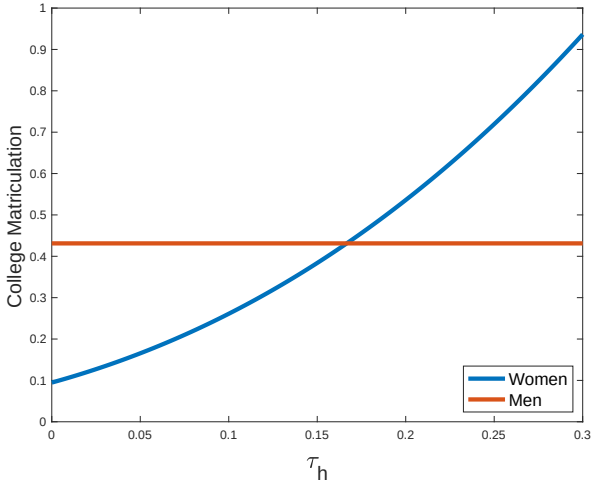
Welfare Gap

$$\text{Welfare} \;=\; (1-\Pi_g)\times V_g^h \;+\; \Pi_g \times \mathbb{E}\left[\, V_g^c(\varphi) \;\middle|\; \text{enrolled in college} \right]$$



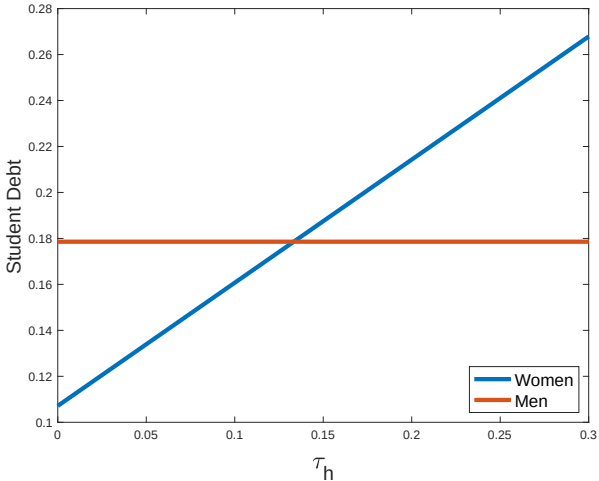
College Matriculation

$$\Pi_g = \left[\frac{(1-\tau_c)w_c - (1-\tau_h)w_h}{p/\chi} \right]^{\theta}$$



Student Debt Gap

$$\mathbb{E}\left[\frac{1}{\varphi}p \,\middle|\, \text{enrolled in college}\right] = \frac{\theta}{1+\theta}\left[(1-\tau_c)w_c - (1-\tau_h)w_h\right]$$



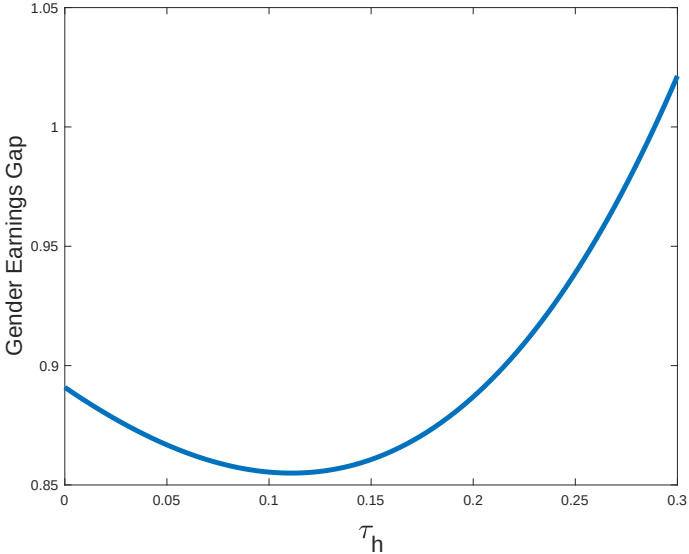
Aggregate Gender Earnings Gap

$$\text{Agg. Gender Earnings Gap} \;=\; \frac{(1-\Pi_w)(1-\tau_h)w_h \;+\; \Pi_w(1-\tau_c)w_c}{(1-\Pi_m)w_h \;+\; \Pi_m w_c}$$

As τ_h increases, the aggregate gender earnings gap

- **Direct:** falls because it reduces earnings of non-college women, $(1-\tau_h)w_h$
- **Compositional:** rises because it increases women's matriculation, Π_w

Aggregate Gender Earnings Gap



College Earnings Premium

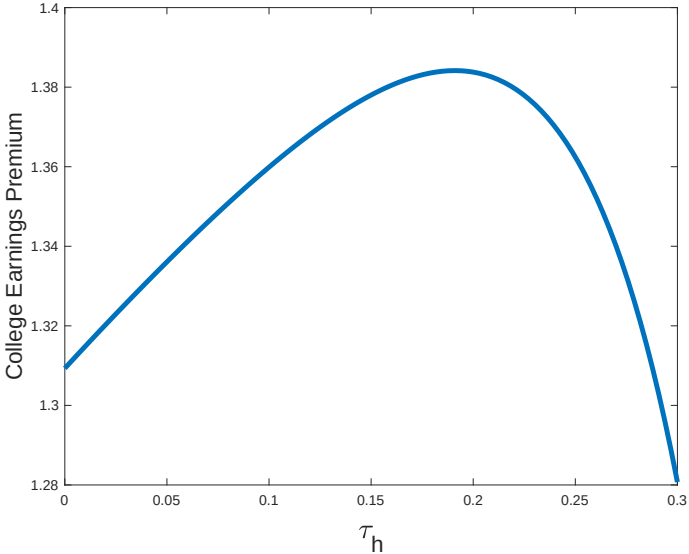
$$\text{College Earnings Premium} \;=\; \frac{\Pi_w^c(1-\tau_c) \;+\; \Pi_m^c}{\Pi_w^h(1-\tau_h) \;+\; \Pi_m^h} \times \frac{w_c}{w_h}$$

$$\text{where} \qquad \Pi_g^c \equiv \frac{\Pi_g}{\Pi_w + \Pi_m} \qquad , \qquad \Pi_g^h \equiv \frac{1-\Pi_g}{1-\Pi_w + 1 - \Pi_m}$$

As τ_h increases, the college earnings premium

- **Direct:** increases by reducing earnings of non-college women, $(1-\tau_h)w_h$
- **Compositional:** decreases because women become greater share of college population Π_w^c and face a gender earnings gap $(1-\tau_c)w_c < w_c$.

Aggregate College Earnings Premium



Additional Facts on College Sorting and Debt

What about the impact of Student Debt?

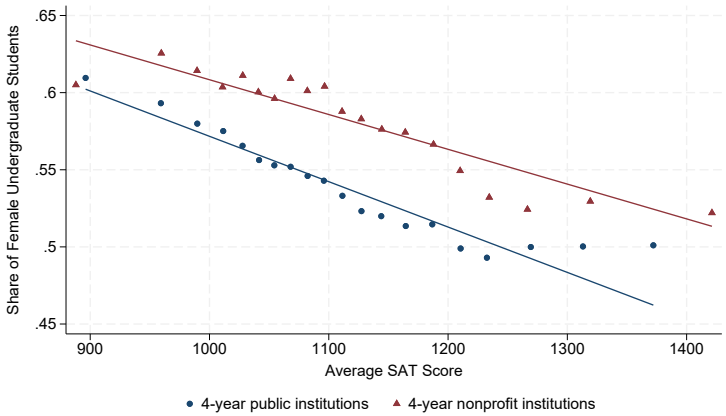
Graduating with more student debt can affect post-college earnings,

- Job Amenities and Initial Earnings (Rothstein Rouse 2011,Mongey Liu 2024)
- Occupation Choice and Returns to Experience (Alon, Cox, and Kim 2024)
- College Major Choice (Hampole 2024)
- Housing and Entrepreneurship (Folch and Mazzone 2022, Morazzoni 2024)
- Job Search and Matching (Herkenhoff, Phillips, Cohen-Cole 2024)

As well as the type of college attended...

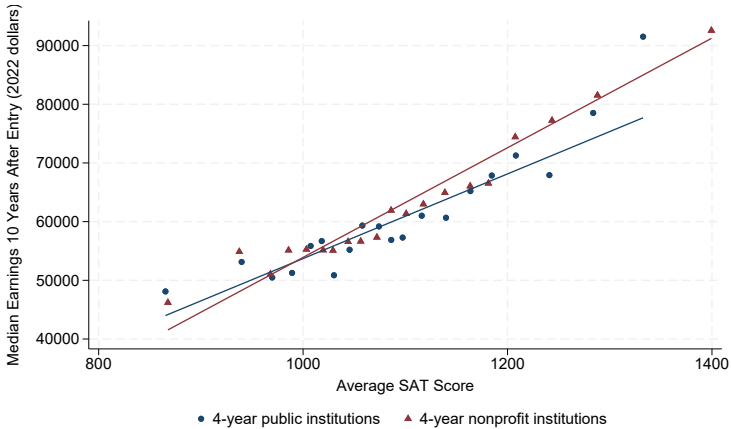
Female Sorting in Higher Education

Women attend lower quality schools on average ...



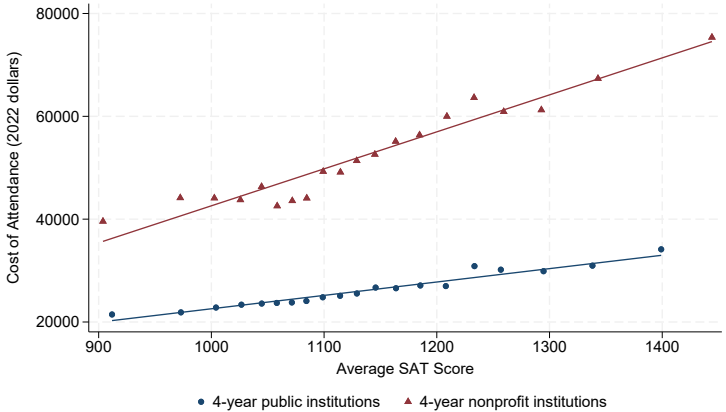
Female Sorting in Higher Education

...resulting in them having lower earnings after graduation



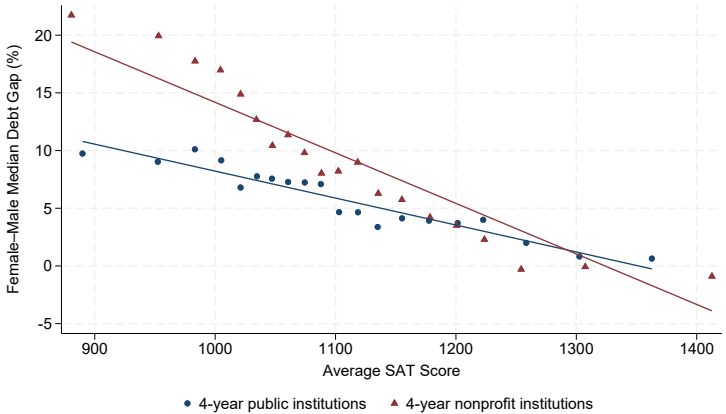
Female Sorting in Higher Education

...and while these schools cost less to attend



Female Sorting in Higher Education

...women still end up with more student debt relative to male peers



Quantitative Model Extensions

Quantitative Model Extensions

- **Endogenous Impact of Student Debt on Earnings** arising from college quality choice, post-grad investment, and financial frictions.

- **Young and Old Workers**
  - Young workers ($t = 1$–4) invest in human capital, repay student debt
  - Old workers ($t = 5$–8) only consumption and labor supply decisions
  - Retirement ($t = 9$) with terminal value assets

- **Production Sector** to endogenize wages w_c, w_h

$$Y = A\left[\zeta H_c^{\eta} + (1-\zeta)H_h^{\eta}\right]^{\frac{1}{\eta}}$$

- **Endogenous Labor Supply** with heterogeneity across men and women

- **Residuals:** unobserved costs τ_s and college preferences κ_g

Old Worker (t = 5–8)

$$V_t^{g,e}(x,a) \;=\; \max_{c,n,a'} \; u(\,c\,,\,n\mid \psi_{g,e}) \;+\; \beta V_{t+1}^{g,e}(x,a')$$

subject to

$$c + a' \;\leq\; (1-\tau_{g,e})wxn + (1+r)a$$

$$a' \;\geq\; -\bar{a}$$

with Greenwood, Hurcowitz, Huffman (1988) preferences

$$u(\,c\,,\,n\mid \psi_{g,e}) \;=\; \frac{1}{1-\rho}\left(\,c \;-\; \psi_{g,e}\,\frac{n^{1+\nu}}{1+\nu}\right)^{1-\rho}$$

$$V_R(a) \;=\; \mathcal{R}u(a) \qquad (t=9)$$

Young Worker ($t = 1$–4)

$$V_t^{g,e}(x, d, a) \;=\; \max_{c,s,n,a'} \;\; u(\, c\,,\; n\,,\; s \mid \psi_{g,e}) \;\;+\;\; \beta\, V_{t+1}^{g,e}(x', d, a')$$

subject to

$$\begin{aligned} c \;+\; a' \;&\leq\; (1 - \tau_{g,e}) w (1 - s) x n \;+\; (1 + r) a \;-\; \Delta(d, t) \\ x' \;&=\; B x s^{\alpha} \;+\; (1 - \delta) x \\ a' \;&\geq\; -\bar{a} \end{aligned}$$

with Greenwood, Hurcowitz, Huffman (1988) preferences

$$u(\, c\,,\; n\,,\; s \mid \psi_{g,e}) \;=\; \frac{1}{1 - \rho} \left(c \;-\; \psi_{g,e} (1 - s)\, \frac{n^{1+\nu}}{1 + \nu} \right)^{1-\rho}$$

Higher Education: School Choice

Intensive Margin: Choose college quality (q):

$$q^* \;=\; \text{argmax}_q \;\; V_1^{c,g}(a^*, x^*, d^*)$$

where

$$\text{cost}(q) \;=\; \frac{1+\tau_{g,s}}{\varphi}\,(p+\omega q)$$

$$\begin{aligned} a^*(a_0) &= \max\Big\{0,\; a_0 - (1+\tau_{g,s})(p+\omega q^*)\Big\} \\ d^*(a_0) &= \min\Big\{0,\; a_0 - (1+\tau_{g,s})(p+\omega q^*)\Big\} \\ x^*(a_0) &= x_0 \cdot (q^*)^{\gamma} + 1 \end{aligned}$$

Higher Education: Attendance choice

Extensive Margin: College matriculation at $t = 0$

$$V^g(a_0) \;=\; \max\Big\{\; V^{c,g}(x^*, d^*, a^*) + \kappa^g + \epsilon^c \;,\;\; V^{h,g}(1, 0, a_0) + \epsilon^h \;\Big\}$$

where ϵ are i.i.d. extreme value shocks with variance σ^2.

Production Technology

Production uses skilled and unskilled labor

$$Y = A\left[\zeta H_c^{\eta} + (1-\zeta)H_h^{\eta}\right]^{\frac{1}{\eta}}$$

where H_e is total effective hours of education group e,

$$H_e = \sum_g \left(\sum_{t=1}^{8} \mathbb{E}_t^{g,e}\left[(1-\tau_{g,e})(1-s^*)x^* n^*\right] \right)$$

Aggregate Implications

Aggregates now depends on changes in matriculation and earnings

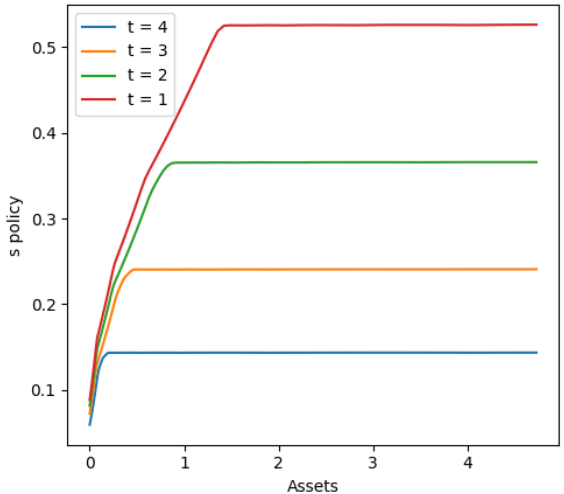
$$\text{Agg. Gender Earnings Gap} \quad = \quad \frac{(1-\Pi_w)\,Y_t^{w,h} \;+\; \Pi_w\,Y_t^{w,c}}{(1-\Pi_m)\,Y_t^{m,h} \;+\; \Pi_m\,Y_t^{m,c}}$$

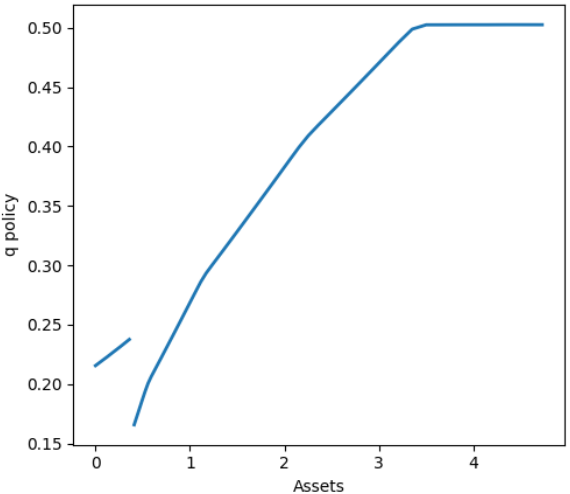
$$\text{College Earnings Premium} \quad = \quad \frac{\Pi_w^c\,Y_t^{w,c} \;+\; (1-\Pi_w^c)\,Y_t^{m,c}}{\Pi_w^h\,Y_t^{w,h} \;+\; (1-\Pi_w^h)\,Y_t^{m,h}}$$

where earnings vary endogenously with education, gender, student debt and age

$$Y^e = \mathbb{E}\left[(1-\tau_e)w_e(1-s^*)x^*n^* \,\middle|\, \text{education level } e^*\right]$$

Human Capital Policy Functions





Counterfactual Exercises

Objective: Quantify the impact of τ on the (i) twin education gaps, (ii) college earnings premium, and (iii) aggregate gender earnings gap.

Strategy: Structural estimation of τ_h,τ_c using stylized facts and causal evidence.

Today:

- Set $\tau_h = \tau_c = 0$

Work in Progress: Transition dynamics, 1980–2025.

Calibration Strategy

Calibration Strategy: Steady State Targets

Match Key Moments by Education and Gender:

- Earnings (ζ, $\tau_{g,e}$)
- Labor Supply ($\psi_{g,e}$)
- Matriculation (κ^g)
- Student Debt (p, $\tau_{g,s}$)

Match Heterogeneity in College Quality:

- Gender sorting across college types (ω)
- Earnings gradient of college quality (γ)

Calibration Strategy: Steady State Targets

Match Moments of Asset Distribution

- Initialize households to match empirical asset and student debt distributions in SCF.
- Average assets to income (β)
- Average assets at retirement ($\mathcal{R}$)

Calibration Strategy: Causal Evidence on Key Parameters

How do changes in outside options effect college matriculation?

- New evidence from Fracking shocks Fracking Regression
- Pins down shape of education taste shocks (σ)

$$\mathbb{E}\left[\,\Pi(a_0, x_0) \mid t = 1,\ +1\%\ \text{HS earnings}\,\right] - \mathbb{E}\left[\,\Pi(a_0, x_0) \mid t = 1,\ \text{SS}\,\right]$$

How does greater college debt affect post-college earnings?

- Impact of debt on initial earnings and RTE (Alon et al 2025, Liu and Mongey 2024)
- Pins down α of human capital technology

$$\mathbb{E}\left[(1-\tau_{g,c})w_e(1-s^*)x^*n^* \mid t = 1,\ \text{+1\% student debt}\right]$$
$$-\,\mathbb{E}\left[(1-\tau_{g,c})w_e(1-s^*)x^*n^* \mid t = 1,\ \text{SS}\right]$$

Calibration Strategy: Causal Evidence on Key Parameters

Outside options $\rightarrow$ matriculation (pins down σ)

$$X\% \;=\; \mathbb{E}\left[\,\Pi(a_0, x_0) \mid \kappa = 1,\ +1\%\ \text{HS earnings}\,\right] - \mathbb{E}\left[\,\Pi(a_0, x_0) \mid \kappa = 1,\ \text{SS}\,\right]$$

Student debt $\rightarrow$ earnings (pins down α)

$$\begin{aligned} X\% \;= \mathbb{E}\left[(1-\tau_{g,e})w_e(1-s^*)x^*n^* \mid \kappa = 1,\ \text{treat}\right] \\ - \mathbb{E}\left[(1-\tau_{g,e})w_e(1-s^*)x^*n^* \mid \kappa = 1,\ \text{SS}\right] \end{aligned}$$

Preliminary Results

I. Counterfactuals: Impact of τ_h and τ_c

	Data	Model	$\tau_h = 0$	$\tau_c = 0$	$\tau_X = 0$
Matriculation Gap	1.28	1.28	1.11	1.35	1.17
Student Debt Gap	1.29	1.29	1.29	1.30	1.30
Labor Supply Gap	0.95	0.95	0.99	0.96	1.00
Gender Earnings Gap	0.86	0.86	0.91	0.93	0.98
College Earnings Premium	1.83	1.83	1.77	1.86	1.79

II. Counterfactuals: Contribution of Exogenous Gender Gaps

	Data	Model	$\tau_X = 0$	$\kappa_w = \kappa_m$	$\psi_w = \psi_m$
Matriculation Gap	1.28	1.28	1.17	0.83	1.76
Student Debt Gap	1.29	1.29	1.30	1.30	1.34
Labor Supply Gap	0.95	0.95	1.00	0.92	0.91
Gender Earnings Gap	0.86	0.86	0.98	0.76	0.93
College Earnings Premium	1.83	1.83	1.79	1.60	1.42

Conclusion

- Analyze the emergence of seemingly paradoxical twin gender gap
- Document significant role of relative differences in outside options
- Quantify implications for the composition and earnings of higher education
- Quantify implications for measurement and dynamics of gender inequality

Work in Progress:

- Transition Path dynamics for gender gap, college premium, and matriculation.
- Micro evidence on sources of student debt gaps and financial consequences (e.g. other forms of credit, bankruptcy, delinquency, repayment, etc)

The Data: Linked FAFSA and Credit Records

Administrative Data from California (2006–2023)

- **FAFSA Records:** Universe of federal aid applicants (2006–2015)
  - Detailed parent and student information: income, dependency, school lists
  - Enables parent-child link via SSNs (for dependent students)
- **Credit Bureau Records:** Quarterly tradeline data (2004–2023)
  - Credit cards, student loans, auto loans, mortgages, HELOCs
  - Credit scores, delinquencies, and bankruptcies
  - Parents' and students' financial trajectories tracked before and after college
- **Key Advantage:** Large scale (2 million+) linking parents and kids
- **Sample:** All FAFSA filers in California; 73% attend college

Appendix

Causal Evidence on Outside Options and Matriculation

Table: Estimates Log High-school Income on Enrollment

	(1) OLS	(2) OLS	(3) OLS	(4) IV	(5) IV	(6) IV
Log HS Income	-0.00205	0.00403	-0.00491	-0.184***	-0.193***	-0.114***
	(0.00345)	(0.00551)	(0.00471)	(0.0134)	(0.0168)	(0.0147)
Observations	17470	8735	8735	17470	8735	8735
F-stat				29.134	41.52	158.69
Sample	All	Men	Women	All	Men	Women

Standard errors in parentheses. SEs clustered by county and year. All regressions include county and year fixed effects, as well as controls for share nonwhite, share less than high school education, share more than college education, share younger than 25 and share older than 55.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Back